Handoff 05: assembly and bring-up

For: Matt at the bench, with Claude Code in VS Code following along. Claude reads the debug log, checks values against `docs/verification/VERIFY.md` and records them there. Allow about 3 h for the first lamp.

You need:

- **Tools:** fine-tip iron, hot plate and/or hot air, flux, IPA, tweezers, a 10× loupe, a multimeter with diode test, and a bench supply with a current limit (or a spare charged cell plus the meter).
- **Stencils and jig:** main board 0.12 mm; LED board **0.10 mm bottom** (WQFN-20) and 0.12 mm top; plus the printed `jig_led_board` (PETG, from `print/`). If the LED board's bottom face was fab-assembled (handoff 01), skip §2.1.
- **Fitment views:** `docs/design/fitment/`, rendered by `cad/tools/fitment.py`. They show where every part sits: the exploded pack and moon head, and sections through the pack and the head.
- **Drawings:** `fab/main-board/v0.1/` and `fab/led-board/v0.2/`: `assembly_top.pdf` / `assembly_bottom.pdf`. These have values on the Fab layer and J4 crossed out as DNP. The zoomed views are easier to read: `renders/*_assembly_*.png` in the same folders.
- **Off-board items** (`bom/other_purchases.csv`): Kapton tape, PTFE sleeve, heat-shrink, CA gel, and neutral-cure silicone (or E6000).

1. Main board (all parts on top)

1. Inspect the bare board. There must be no shorts between VBUS / BAT_RAW / VSYS / VLED / GND: meter TP1, TP2, TP3 and TP7 to TP4 (GND).

2. Stencil and paste, then place the parts per the assembly drawing:

   - the passives;
   - SOT-23s Q1–Q4, checking the orientation dots;
   - U1 MCP73831 and U2 ATtiny3216;
   - D1/D3 with the cathode bar to the silk bar, and the red 0603 D2 with its cathode mark to pad 1;
   - J1 USB-C, SW2–4 tact switches, J2 PH and J3 SH.

   Reflow (SAC305, or leaded paste for a lower peak). Then solder J1's four shell slots through the holes with the iron.

3. **Fit SW1 (PCM12) by iron after the reflow.** It is rated 255 °C max, so it must not go through a hot plate (round2 R2-D13). **Do not fit J4**: hold a header in the holes when programming. A soldered header would point at the pouch (round1 B-2).

4. Clean off the flux. Under the loupe, check U2's pins and the USB-C pads for bridges.

5. Run VERIFY **E1–E4** now (bench supply on J2, no cell, no LED board). E2 finds which knob direction is ON: mark it.

6. Flash the firmware: handoff 04, steps 1–5. With no LED board fitted, the lamp just retries quietly. The debug log shows `T=-99` and then the 30 % fail-safe cap, which is expected.

2. LED board v0.2 (18 mm; bottom first, then top)

1. **Bottom side** (0.10 mm stencil): U1 LP5012 (WQFN-20, 3 × 3 mm, the package's pin-1 dot to the silk pin-1 triangle; the exposed pad is GND), U2 MCP9800A0 (SOT-23-5), R1 2.94 kΩ, C1 and C4 1 µF, C3 100 nF (all 0402), C2 10 µF (0603). Reflow on the hot plate. A WQFN's joints are hidden: look at its sides under the loupe for paste balls or bridges, then meter V–G, D–G, C–G and D–C on the cable pads: none may be a short. Put a **Kapton disc over U1** afterwards: it faces the gooseneck's metal tube end (round2 R2-D3).

2. **Top side.** Put the board in `jig_led_board`, LEDs up. The jig supports only the part-free edge band, so the bottom parts hang free and the board top is flush with the jig top. Stencil (or paste dots), then place:

   - **8 × white 2835, radial, large cathode pad toward the board centre.** The silk letter next to each one says which: **W = 2700 K** (D1, D3, D5, D7) and **C = 5000 K** (D2, D4, D6, D8).
   - **1 × Würth RGB (D9) at the centre.** The package's polarity mark (chamfer/dot) is **pin 3, the red cathode**: put it on the **silk chamfer**. Pin 1, the common anode, then lands on the pad marked "1". Don't put the package's mark on the "1" (round2 R2-D4).
   - **R2 2.4 kΩ (0402)** in the outer gap between D1 and D2: the LP5012's EN pull-up. Without it the driver stays in shutdown and every LED stays dark.

   Reflow with hot air, ≤ 260 °C and ≤ 10 s at peak. Or lift the board off the jig onto two metal spacers under the edge band and use the hot plate. The jig is PETG, so it is for stencilling and hot air, not a 240 °C plate. The 2835s have bottom-only terminations, so an iron can't solder them. The bottom parts will re-melt; surface tension holds them.

3. Diode-test each LED (VERIFY E16): + probe on pad V (VLED, all anodes), − probe on the exposed inner end of its cathode pad (it sticks out 0.25 mm past the body; the vias are tented). Each one should glow faintly in the right colour; a dark one is reversed or open on top. Green and blue need about 2.6 V to glow, so a meter whose diode test gives less can only check them at first light. Opens under U1 show up only at first light (§5): each mode lights a known set (2700 K = the four W, 5000 K = the four C, red/green/blue = one die each).

3. Cable and neck (round1 M-7: abrasion and strain relief)

Qwiic cable colours: **black = GND, red = VLED, blue = SDA, yellow = SCL**. The LED board pads are labelled **G V D C** on its bottom silk.

1. Deburr the inside of both gooseneck ends with a needle file. Self-tap the gooseneck's lower end into `pack_front`'s 9.3 mm socket by hand, before any cable is fitted, so nothing twists. A drop of CA gel on the thread is optional. **No anaerobic thread-locker**: it crazes PETG (round1 m-7).
2. **Moon end, before any cable** (round3 R3-M4/M5): run the jam nut, then the washer, onto the gooseneck's top thread. Drop the second nut into the cup's hex socket and screw the cup (with that nut) onto the thread, turning the cup, until the thread end is flush with the socket nut's top (≤ 0.5 mm above it, VERIFY M12; the thread needs ≥ 9 mm, M1). Then a drop of CA gel on the thread and run the jam nut up against the washer and the floor,

holding the cup: the floor is now clamped between the two nuts. If the thread stands proud, cut or file it now and blow out the filings, not after the cable is in.
3. Cut the Qwiic cable to about 200 mm, keeping the JST-SH end. Slide the PTFE sleeve over the 4 wires. Feed the bare end up through the gooseneck from the pack side and out through the socket nut. Put 15 mm of heat-shrink over the wires where they leave the pack end; at the moon end the heat-shrink (or the PTFE sleeve) stays inside the bore, flush with the thread end (there are only 4 mm above it). The wires leave the bore above the nut, into the 4.5 mm space under the board.
4. Solder G / V / D / C on the LED board (the 2 × 2 pads on its bottom). Cover the joints and 5 mm of wire with neutral-cure silicone, and keep the blob **inside r ≈ 8 mm** (the pads end at r 7.7; the cup's ledge starts at r 8.35): silicone on the ledge would tilt the board. Let it cure. This is the strain relief.
5. Lay the LED board in the cup's pocket (Ø18.4, LEDs up; its top stands 0.4 mm proud of the rim land), then put the PETG `moon_ring` on its edge band, chamfer up. Fit the moon: slide its skirt down over the cup with the three entry slots on the cup's lugs, then turn it **clockwise seen from the top, like a jar lid**, 30° until it stops (the grooves run one way only). The cam pulls the moon down and the ring flexes into the gap above the land: that is the clamp's spring. The board must not rattle (VERIFY M5). **Whenever you turn the moon, hold the cup band below the skirt**, so the twist never reaches the neck thread or the cable.

4. Pack

1. Put Kapton tape over the top face of the pouch, and a patch on the PCB back under U1, D3 and Q1–Q3.
2. **VERIFY E1 again on the real cell: + must be on J2 pin 1.** From the back of the PCB, pass the cell's plug up through the 6.4 × 5.0 mm slot beside J2, mate it with J2, and let it lie flat over the slot. Lay the slack lead flat in the bay's end gap, clear of the two screw heads (VERIFY M13).
3. Fit the button caps into the front shell. **Check SW1 is OFF**, then plug the JST-SH into J3: J3 is mated once, here, and never with VLED live (plugging it live can reset the MCU at a low battery: SPICE report D2). Leave a 10–15 mm service loop, then fix the cable to the gooseneck bore entrance in the neck block with a dab of silicone. A pull on the neck is then taken there and never reaches the JST-SH, which has no latch. Screw the PCB onto the bosses with 2 × M2×6.
4. Battery into the bay. Hook the back-plate tabs into the slots, then fit 2 × M2×12 through the clip root and the plate into the front shell.

5. First light and checks

1. Switch ON with no USB. The lamp should fade into Soft white 3000 K at level 10. Run through the modes, UP/DOWN and hold-to-save (two dips).
2. Debug build (handoff 04 §5): run VERIFY **F2–F5** and **E7–E17**. E13 is the scope check of the LED-board soft turn-on.
3. Before regular use, run at least **T3** (4000 K full for 2 h with thermocouples, or a contact thermometer on the moon skirt) and **T6** (the thermostat). Record the results in VERIFY.md and send them to Claude, who re-fits the derating if the board-to-PLA offset is not about 14 K.
4. Light and colour: VERIFY **O1–O7** as equipment allows. A phone lux app is good enough for O3.

6. User notes (for whoever uses the lamp)

- Charge from a 5 V USB-C charger rated 1 A or more. On a weak source (an unpowered hub, a long thin cable) the lamp can show a low-battery warning while plugged in; it is harmless (SPICE report D3).
- Charge at room temperature (the charger has no cell-temperature sensing). The red LED means charging. It flickers when no battery is connected, which is normal.
- The side switch is the only off. Switch off when you put the lamp down.
- After a "battery empty" fade-out, plug in USB. The lamp comes back on by itself after a few seconds.
- If the pack ever bulges, stop using it and replace the cell.
- Never power the lamp from a programmer or a USB-UART adapter.

Done when

One lamp is assembled and every E, F and T3/T6 item has a measured value in VERIFY.md. Anything outside its pass band is raised with Claude before the second lamp is built.